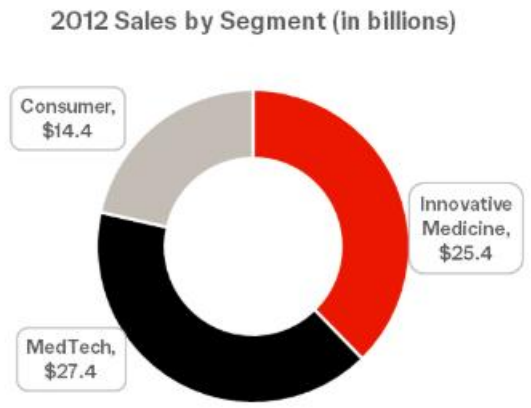
Johnson&Johnson

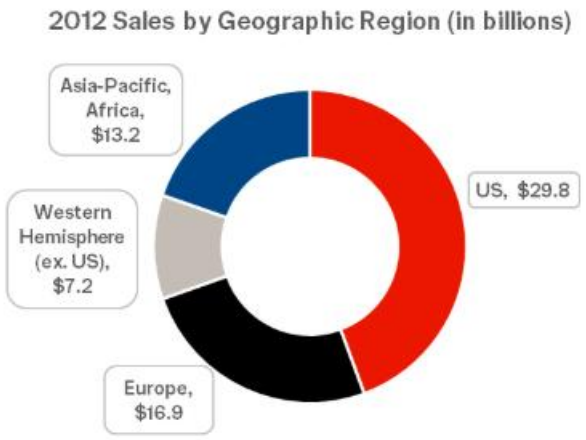
2026 University Case Competition Draft

Aegis & Aegis Background

Aegis & Aegis (A&A) is a global company engaged in the research and development, manufacture, and sale of a broad range of products in the healthcare field. The company was founded in 1886 and is headquartered in New Brunswick, New Jersey. With a strong focus in innovation and quality, the company is a leader in the industry and is one of the 2012 Fortune 50 companies with over 128,000 employees and more than 250 operating companies worldwide. A&A is grounded in the strong principles of its Credo which guides decision making (see Appendix 3 for more information).

A&A has three business segments, Medical Devices & Diagnostics, Innovative Medicine, and Consumer Health.** In 2012, A&A reported sales of $67.2 Billion with 41% of the business in the Medical Devices & Diagnostics segment, 38% in the Innovative Medicine segment, and 21% in the Consumer segment.





Medical Devices & Diagnostics is a key component of A&A's business, and the company continues to seek opportunities to accelerate the segment's growth. Through novel internal research and strategic external partnerships, the company intends to strengthen its portfolio to offer a diverse set of solutions for patient care. By prioritizing patients at its core and using the values outlined in the Credo, A&A continues to be an innovator in the industry to provide the best patient outcome.

***The current state of Aegis & Aegis in 2025 looks much different than it did in 2012. A&A's consumer health business was spun off in 2023 to become the public company, Zeus. Given the case decision is based on a time prior to the business evolving, company background outside of what is shared in the case should not be used.*

The Challenge

Your team is tasked with deciding the best strategy to ensure a successful expansion of A&A's wound closure portfolio. Your strategy should outline how to best grow the company and its expertise, while also maintaining the strong values that A&A has. In creating your business plan, you must analyze the

following business strategies and make a robust business plan making decisions on how to proceed with the following:

1. Research & Development (R&D) – Determine if A&A should purchase intellectual property (IP) or internally develop spiral and symmetrical sutures.
2. Pricing – Recommend pricing for patient sutures.
3. Marketing – Decide on an approach for marketing engagement with stakeholders.
4. Supply Chain – Decide to outsource, expand, or build a plant for suture manufacturing.
5. Human Capital Strategy – Determine to hire contractors or full-time workforce for supply chain.

This challenge will require your team to present your business plan and justify your decision points with strong insights to senior management. To develop the best strategy, you will need to effectively collaborate with your team, use your technical skills and think critically to gain a thorough understanding of the company and marketplace it operates in. A well-thought-out business plan must include detailed financial information that demonstrates how your strategy will drive growth for Aegis & Aegis's Medical Devices & Diagnostics segment. While making decisions it is also important to align your strategy with the values outlined in the Credo.

Your team will need to deliver a 20-minute presentation that incorporates the following requirements. Please note that your presentation may not exceed the 20-minute time allotment, as this will be followed by a 10-minute question & answer session (total 30 minutes). Your team may choose to organize the below topics in any order as long as all are covered. (See Appendix 1 for more detail.) **Please note that outside research may be used to aid in the SWOT/Competitor analysis ONLY; no other parts of the analysis may rely on outside research.**

a. Market Analysis (SWOT) - **10%**
b. Detailed Financial Metrics (NTS, NPV, IRR, additional financial information) - **25%**
c. Business Decisions - **40%**
d. Quality of Presentation including Q&A - **25%**

Be prepared for senior management to challenge your decision points and assumptions at the end of your presentation. The quality of your presentation and your ability to answer their questions will affect your team's final score. Any uncertainties or gaps in your assessment will factor in senior management's decision on whether to pursue your strategy.

Good Luck!

Business Area Overview

Medical Devices & Diagnostics Background

Aegis & Aegis's Medical Devices & Diagnostics segment focuses on Cardiovascular, Vision, Orthopedics, and Surgery among others. These segments account for $27.4 billion in revenue for A&A. The company's mission is to help save lives and create a future where healthcare solutions are smarter, less invasive, and more patient centered. Specifically, Medical Devices & Diagnostics has a goal to provide global access to safe and affordable surgical services for all.

While the overall market growth has slowed, A&A has been focusing on building market leadership positions including currently holding No. 1 or No. 2 positions in 80 percent of key platforms.

Wound Closure Overview

A&A's Surgery business is the current market leader in the non-barbed suture market. The total wound closure market is growing globally, and A&A is interested in expanding its suture portfolio to include barbed sutures. The company is a major supplier of sutures to hospitals and surgical centers. A&A currently has one of the suture manufacturing plants located in Cornelia, Georgia. Customers value the quality and reliability that A&A has consistently supplied, improved patient outcomes, and efficiency these sutures provide to the surgeons who use them daily.

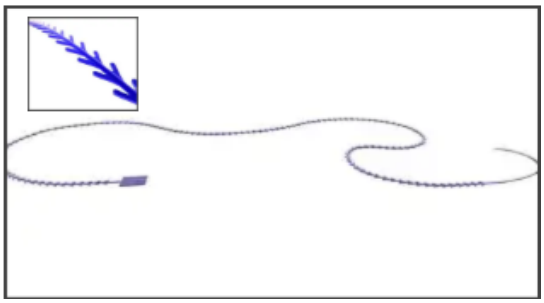
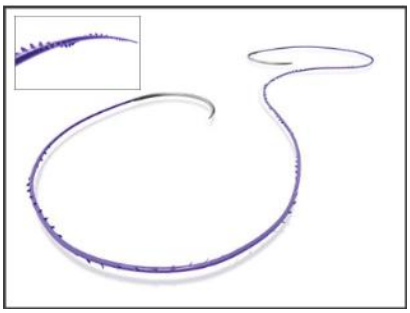
A&A Surgery faces some challenges within the suture space, including:

- Competition from Alternatives: Surgical Staples and Tissue Adhesives
- Regulatory Hurdles: Complex & Costly Approval Process, Regional Variations in Regulations
- Cost and Affordability: Pricing Pressures from Competition, Cost of Raw Materials

Currently, there is a shift from non-barbed to barbed suture adoption. Previously, the affordability and the wide-spread surgical training anchored to non-barbed sutures had hindered the growth of new products entering the market. But, with recent advancements in robotic surgery, those barriers are not as prevalent.

The FDA approval of the Talos Surgical System in the year 2000 led to widespread adoption of robotic surgeries. By 2010, robotic surgeries were commonplace, and a new market emerged to improve inefficiencies in the operating room. Barbed sutures offer a solution, allowing for shorter, less invasive surgeries. The knotless design of these sutures makes them water-tight, decreasing the risk of post-operative infection.

Exhibit A:

Symmetrical	Spiral
Symmetrical sutures are a type of barbed suture arranged in a mirror image pattern on either side of the suture axis. The symmetrical suture is a widely used affordable option, but the design only allows for unidirectional use.	Spiral sutures are a type of barbed suture arranged in a helical pattern around the axis with an adjustable fixation loop at the end to eliminate the need for knots during minimally invasive surgeries. The spiral structure allows for the suture to be unidirectional or bidirectional.
	

Competitive Landscape

Crius (Poseidon)

Crius was founded in 2007 as a spin-off from Themis International, a company that previously operated in three main sectors: Themis Healthcare, Themis Electronics Ltd, and Themis International Ltd. (Fire Safety & Security). Headquartered in Dublin, Ireland, Crius has quickly established itself as a leader in the global medical equipment manufacturing field.

By 2010, Crius's gained FDA approval for its flagship product, Poseidon, a wound closure device that features a dual angle cut on its suture. The shallower depth of the suture can improve its effectiveness in surgical procedures.

In 2012, Crius reported revenues of approximately $11.9 billion, with 55% of sales coming from the U.S. and 45% from international markets. The company employed around 43,400 people who primarily focused on their three sectors — Medical Devices, Pharmaceuticals, and Medical Supplies. The Medical Devices sector, which includes their wound closure devices, generates the majority of revenue at around 68%.

Athena Pharmaceuticals, Inc. (Kronos)

Athena Pharmaceuticals, Inc. is a global surgical solutions company headquartered in Vancouver, British Columbia, Canada, that provides Medical Device Technologies and Licensed Technologies through three segments: oncology, ophthalmology, and wound closure. They employ 1,364 people across 8 different countries.

Athena first obtained a patent for its self-anchoring symmetrical design of suture, Kronos, in 1999 and subsequently gained FDA approval by 2005. It was then rolled out to Europe in October of 2007. Its Kronos product portfolio includes both bidirectional barbed sutures—designed to change direction midway—and unidirectional barbed sutures with an adjustable loop at one end, offering advanced options for surgical procedures.

In 2012, Athena generated over $221.3 million in revenue with 62% coming from U.S. markets and 38% derived from international markets. Although its product portfolio is not as large as competitors, its differentiated wound closure devices and promising revenue growth allow Athena to compete with the larger players.

Other Competitors

Additionally, a handful of companies have shown recent or promising innovations in the suture space. Hades Medical, a small private company based out of Flagstaff, AZ, released its Preclude AJA Nymphs Substitute in Japan in 2008. This suture substitute provides a nonwatertight underlay graft for craniotomies (NIH). Apollo & Nyx, a large publicly traded company, invested in their GAIA product line for negative pressure wound therapy. Their single use negative pressure wound therapy system is an alternative to traditional wound dressing and is intended to treat complex open wounds and high-risk surgical incisions. With new competitors entering the market, A&A's investment in this space will be pivotal to maintaining its market share position.

Decision Points

The decision points below form the basis of your team's strategy. You will input these into the Excel model and present your findings to management. See each section below for more information.

R&D Investment Strategy
Option 1: Purchase IP of Spiral & Symmetrical **Option 2:** Internally Develop Spiral & Symmetrical **Option 3:** Purchase IP of Spiral; Internally Develop Symmetrical **Option 4:** Internally Develop Spiral; Purchase IP of Symmetrical

Supply Chain Strategy
Option 1: Outsource **Option 2**: Plant Extension **Option 3**: New Plant Construction

Pricing Strategy
Option 1: Low **Option 2**: Medium **Option 3**: High

Marketing Strategy
Option 1: Bronze **Option 2**: Silver **Option 3**: Gold

Human Capital Strategy
Option 1: Hire contractors **Option 2**: Hire full-time employees

R&D Strategy

Aegis & Aegis wants to upgrade its wound closure portfolio immediately with a best-in-class symmetrical suture to retain market share against its competitors in the industry. There is also an opportunity to be first to market with a spiral suture that has bidirectional capabilities. However, a spiral suture is not on the market yet and would be a premium product with limited buyers until production and adoption are scaled.

There are competitors developing both symmetrical and spiral sutures; however, no spiral products have entered the active market. Investing internally to develop spiral sutures has both a greater risk and

reward since it requires upfront investment, but the company that comes to market first inherits greater market share. When purchasing IP, market share is guaranteed, but the product will not be first to market, therefore you will have to compete with existing products.

The team will need to decide between investing internally in the research and development of their own symmetrical and spiral products, investing in the intellectual property (IP) of an existing symmetrical product or waiting to invest in a spiral product until it comes to market from a competitor. Please use Exhibit A to reference the benefits of each suture.

Product	R&D Strategy	Cash Flow	Market Share
Spiral	Purchase IP	More CapEx; Later to Market	Lower
	Internally Develop	More R&D; Earlier to Market	Higher
Symmetrical	Purchase IP	More CapEx; Earlier to Market	Higher
	Internally Develop	More R&D; Later to Market	Lower

Pricing Strategy

Your team must choose one of the three pricing options below for Aegis & Aegis to charge for the sutures, keeping in mind the impact it will have on the overall profitability of the business and affordability for its target market.

1. **Low:** This is the lowest dollar value Aegis & Aegis can charge for this product to maintain its current business structure. This pricing makes the product affordable and caters to every customer. This price is below the market average and is very competitive in the current market.

2. **Medium:** This dollar value will sustain Aegis & Aegis's current business structure while allowing for minimal redistribution of profit among the business but is only affordable for 80% of the target market. This pricing is the market average and directly competes with some of the largest competitors.

3. **High:** This dollar value will provide plentiful profits to sustain the current business structure and allow for increased investment in other areas of the business. This is only affordable for 65% of the target market. This price is above competitors in the wound closure market but can be justified by the quality of the product.

Marketing Strategy

To have a successful product expansion and launch, Aegis & Aegis will need to identify and effectively reach their target market. Choose between the three investment options below and be prepared to defend its impact on the overall margin, potential effectiveness of strategy, and its relationship to other decision points.

1. **Bronze:** A&A will invest in identifying its target audience and effectively reaching it using digital advertisements. This will allow them to do minimal market research on which hospitals would have an interest in deploying a new type of suture. A&A will release a series of digital advertisements in the form of emails to its existing client list as well as other clients who may be interested.

2. **Silver:** In addition to investments in the Bronze Tier Marketing strategy, A&A will invest in paper materials such as posters and pamphlets. They will be able to accurately gauge new potential clients for an innovative wound closure product, while effectively reaching them through pamphlets that provide information on the benefits the new product has to offer.

3. **Gold:** In addition to investments in the Bronze & Silver Tier Marketing strategy, A&A will invest in attending medical innovation conferences worldwide and providing in-person demonstrations to potential new clients. This will require A&A to send medical sales representatives to gauge interest and build lasting relationships with hospitals and surgeons.

Supply Chain Strategy

A&A is evaluating three primary manufacturing strategies for production of its new barbed sutures: partnering with an external manufacturer, expanding existing facilities, or constructing a new plant. Opting for in-house production through plant expansion or a new facility offers A&A complete control and oversight over the manufacturing process, ensuring strict quality management and protection of proprietary technologies. Outsourcing allows for collaboration with an experienced external manufacturer, potentially reducing costs and increasing production flexibility. Each approach presents its own set of advantages and challenges, which are critical to consider in alignment with A&A's strategic objectives and operational requirements. Senior leadership must decide between three options for manufacturing sutures:

1. **Outsourcing Manufacturing:** A&A would partner with an external contracted manufacturer to produce the new barbed suture. This approach eliminates the need for internal hiring and labor management (no Human Capital Strategy), as the external manufacturer would handle all aspects of production. By outsourcing, A&A could significantly reduce lead times, enabling production to commence much sooner than with the in-house options. As a result, this strategy would have an immediate impact on the P&L, allowing A&A to start generating revenue from the new product without the typical delays associated with capital investment and construction.

2. **Plant Expansion:** A&A would expand its existing manufacturing facility in Cornelia, Georgia, which is one of the plants currently producing surgical sutures and wound closure products. This expansion aims to increase capacity to support the manufacturing of the new barbed suture, as the plant is already operating at full capacity. While this approach leverages existing infrastructure, it requires a significant construction period, with an estimated timeline of approximately three years until operational. The expansion would result in increased CapEx in years 1 & 2, but increased capacity would only begin contributing growth to the company's P&L in year four, resulting in a delayed sales impact compared to Outsourcing. Choosing Plant Expansion will also require a choice for the Human Capital Decision Point.

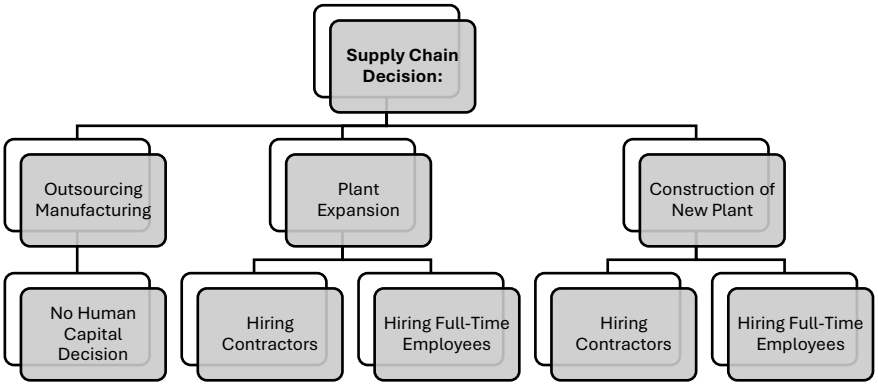
3. **Construction of a New Plant**: A&A would build a new plant with the best technology to specialize in barbed sutures and ensure that the company can keep up with demand. This new plant allows for plenty of space to expand and ramp up production as demand grows. However, this option is the most capital-intensive, involving substantial upfront costs related to design, construction, infrastructure development and labor. Additionally, the timeline for operational readiness is the longest among the options, with an estimated construction and commissioning period of approximately five years. The new plant is expected to become operational in year six,

at which point it will begin contributing growth to the company's P&L. Choosing Construction of a New Plant will require a choice for the Human Capital Decision Point.

With each new product launch, A&A will need to determine how the product can be successfully integrated into its complex global supply chain network. The three supply chain options and some of the impacts are listed below:

Costs	Outsourcing	Plant Expansion	Construction of a New Plant
Initial Investment & CapEx	Lowest CapEx	Moderate CapEx	Highest CapEx
Depreciation	None	Moderate Depreciation Expenses over the Useful Life of Asset	High Depreciation Expense over the Useful Life of Asset
Fixed Operating Costs	Lower	Higher	Higher

When deciding which Supply Chain Manufacturing decision is best for A&A, it is important to leverage how each decision will impact the need for Human Capital as well. Below is a decision tree showing how each will affect the Human Capital decision point:



Human Capital Strategy

To ensure a successful product launch, Aegis & Aegis must structure its human capital strategy for manufacturing to optimize operational efficiency, cost management, and strategic flexibility.

As noted under the previous section, if outsourcing manufacturing is chosen as the Supply Chain Strategy, then there will be no human capital decision to be made as the external manufacturer will handle all aspects of production

If plant expansion or construction of a new plant is chosen as the Supply Chain Strategy, then there will be two primary options are under consideration for human capital strategy: hiring manufacturing contractors or building a full-time manufacturing workforce for supply chain.

The supply chain decision tree above provides a visual of the human capital decision choices available. Below is a chart showing the advantages and disadvantages of both options.

Please note that in the excel file, 'Base' must be selected for human capital strategy if 'Outsource' is chosen for Supply Chain.

Option 1 – Hiring Contractors

Contractors	
Advantages	• Lower up-front cost • Can be engaged on an as-needed basis • Quick onboarding process
Disadvantages	• Lack of in-depth understanding of company culture, processes, and long-term goals • Reduced commitment • Limited loyalty

Option 2 – Hiring Full-Time Employees

Full-Time	
Advantages	• Long-term commitment • Knowledge accumulation • Aligned incentives
Disadvantages	• Higher costs (benefits) • Less flexibility

Appendix 1: Suggested Presentation Content

Your team may choose to organize your presentation in any order, as long as all required information is covered. Please note that not all this content must appear in your presentation but should be available as back-up to support your strategy.

1. SWOT/Risk Analysis
2. Financial Analysis
    a. Financial metrics (Industry Market Growth, Peak Share, Beginning Share, Ending Share, NPV, IRR)
    b. How did your decisions impact the financials?
3. State Assumptions
    a. Decision points selected
    b. Reasons for choice
    c. Reasons for not choosing other options
4. Competitive Analysis
    a. How does A&A product portfolio compete in the market?
5. Sensitivity Analysis of Company Decision
    a. Explore how your key performance metrics vary with changes to your decisions

Appendix 2: Glossary

- **Adjusted Operational Growth Rate** – Company growth rate year over year, excluding all acquisitions and divestitures, as well as currency impacts. It represents pure growth.

- **CAGR (Compound Annual Growth Rate)** – Company's rate of return over a period.

- **Contribution Margin** – Per unit measure of a product's gross operating margin; it is calculated as the product's price minus its total variable costs.

- **Current Assets** – A balance sheet category that represents the value of all assets that are reasonably expected to be converted into cash within one operating cycle in the normal course of business (typically one year). Current assets can include cash, accounts receivable, inventory, marketable securities, prepaid expenses, and other liquid assets that can be readily converted to cash.

- **Current Liabilities** – A balance sheet category that represents the value of all debts or obligations that are due within one operating cycle. Current liabilities can include short-term debt, accounts payable, accrued liabilities, and other debts.

- **Depreciation** – Accounting method to allocate the cost of a tangible asset over its useful life.

- **Discounted Cash Flow** – Financial method of valuing a project, company, or asset using the concepts of time value of money. All future cash flows are estimated and discounted to give their present values ("PV"). The sum of all future cash flows, both incoming and outgoing, is the net present value ("NPV"), which is taken as the value or price of the cash flows in question.

$$DCF = \sum_{n=1}^{T} \frac{CF_n}{(1+r)^n}$$

CF_n = Cash Flow during period n n = Time Period r = Discount Rate

- **Gross Profit** – Company's revenue minus its cost of goods sold. Gross profit is a company's residual profit after selling a product or service and deducting the cost associated with its production.

- **Internal Rate of Return (IRR)** – Discount rate often used in capital budgeting that makes the net present value of all cash flows from a particular project equal to zero. The higher a project's internal rate of return, the more desirable it is to undertake the project.

$$NPV(0) = \sum_{t=0}^{T} \frac{C_t}{(1+IRR)^t} - Initial\ Investment$$

CF_n = Cash Flow during period n t = Time Period
***IRR Formula is to solve for NPV value of zero*

- **Net Trade Sales (NTS)** – Gross sales minus returns and discounts.

- **Net Present Value (NPV)** – The difference between the present value of cash inflows and the present value of cash outflows. NPV is used in capital budgeting to analyze the profitability of an investment or project.

$$NPV = \sum_{t=0}^{T} \frac{C_t}{(1+r)^t} - Initial\ Investment$$

CF_n = Cash Flow during period n n = Time Period r = Discount Rate

- **Profit & Loss (P&L) -** Provides a summary of a company's financial performance over a specific period. It details the revenues generated from sales, the costs and expenses incurred in operations, and the resulting net profit or loss.

- **Return on Investment (ROI)** – A measurement of the efficiency of an investment, calculated by dividing net profit from investment by cost of investment.

- **Working Capital** – Current assets minus current liabilities.

- **Working Capital Ratio** – Current assets divided by current liabilities. This indicates if a company has enough short-term assets to cover its short-term debt.

Appendix 3: Credo

We believe our first responsibility is to the patients, doctors and nurses, to mothers and fathers and all others who use our products and services. In meeting their needs everything we do must be of high quality. We must constantly strive to provide value, reduce our costs and maintain reasonable prices. Customers' orders must be serviced promptly and accurately. Our business partners must have an opportunity to make a fair profit.

We are responsible to our employees who work with us throughout the world. We must provide an inclusive work environment where each person must be considered as an individual. We must respect their diversity and dignity and recognize their merit. They must have a sense of security, fulfillment and purpose in their jobs. Compensation must be fair and adequate, and working conditions clean, orderly and safe. We must support the health and well-being of our employees and help them fulfill their family and other personal responsibilities. Employees must feel free to make suggestions and complaints. There must be equal opportunity for employment, development and advancement for those qualified. We must provide highly capable leaders and their actions must be just and ethical.

We are responsible to the communities in which we live and work and to the world community as well. We must help people be healthier by supporting better access and care in more places around the world. We must be good citizens – support good works and charities, better health and education, and bear our fair share of taxes. We must maintain in good order the property we are privileged to use, protecting the environment and natural resources.

Our final responsibility is to our stockholders. Business must make a sound profit. We must experiment with new ideas. Research must be carried on, innovative programs developed, investments made for the future and mistakes paid for. New equipment must be purchased, new facilities provided, and new products launched. Reserves must be created to provide for adverse times. When we operate according to these principles, the stockholders should realize a fair return.